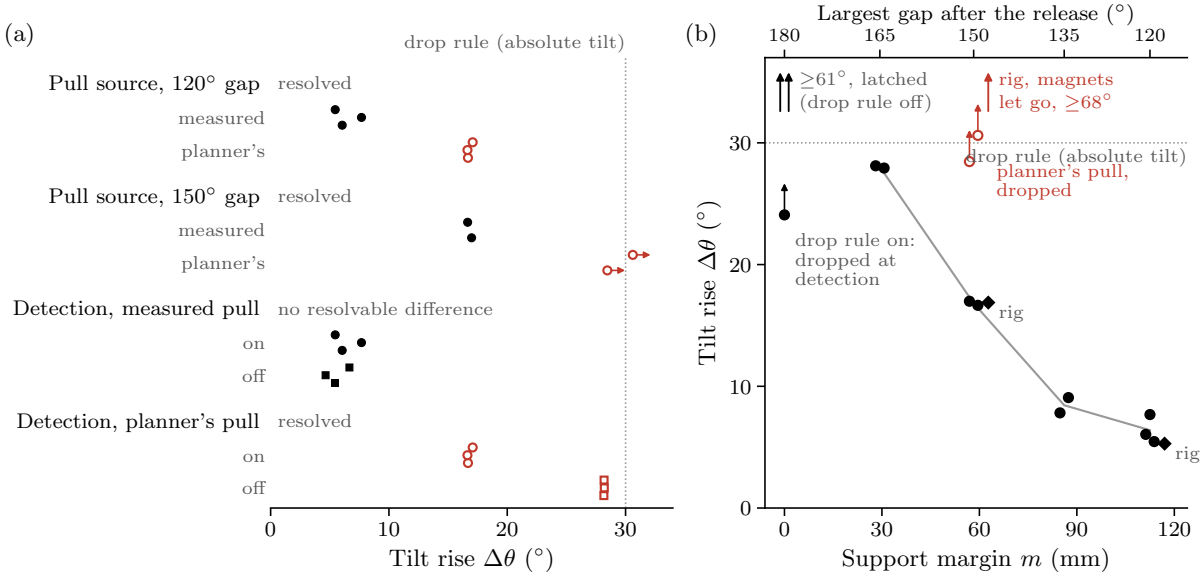

(a)
drop rule (absolute tilt)
Pull source, 120° gap
resolved
measured
planner's
Pull source, 150° gap
resolved
measured
planner's
Detection, measured pull
no resolvable difference
on
off
Detection, planner's pull
resolved
on
off
0
10
20
30
Tilt rise Δθ (°)
(b)
Largest gap after the release (°)
180
165
150
135
120
≥61°, latched
(drop rule off)
rig, magnets
let go, ≥68°
drop rule (absolute tilt)
planner's pull,
dropped
drop rule on:
dropped at
detection
rig
rig
Tilt rise Δθ (°)
0
10
20
30
0
30
60
90
120
Support margin m (mm)